

Problem 1a) (15 points)

Given the image region as shown in Figure 1 and $V = \{1\}$, what is the shortest m-path between p (the pixel at the upper-left corner) and q (the pixel at the bottom-right corner)? Give the length of the path if the path exists.

```

1 0 1 1 0
0 1 0 1 0
0 0 1 1 0
1 1 0 1 0
0 1 1 1 1

```

Step 1: Identify pixels in V

p

1	0	1	1	0
0	1	0	1	0
0	0	1	1	0
1	1	0	1	0
0	1	1	1	1

q

All pixels with value 1 are highlighted. These are the only candidates for the m-path since $V = \{1\}$.

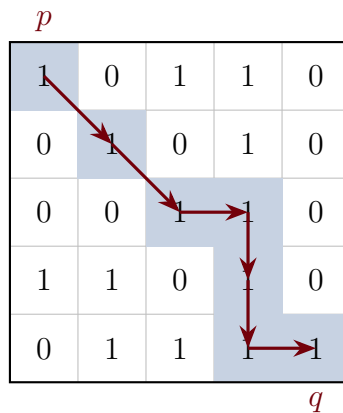
Step 2: Apply m-adjacency rules

Two pixels with values in V are **m-adjacent** if:

- They are 4-adjacent (share an edge), OR
- They are diagonally adjacent AND their shared 4-neighbors contain no pixels with values in V .

The second condition prevents diagonal shortcuts when a 4-connected bridge exists, eliminating path ambiguity.

Step 3: Trace the shortest m-path



Path: $(0, 0) \rightarrow (1, 1) \rightarrow (2, 2) \rightarrow (2, 3) \rightarrow (3, 3) \rightarrow (4, 3) \rightarrow (4, 4)$

Diagonal move justifications:

- $(0, 0) \rightarrow (1, 1)$: Shared 4-neighbors are $(0, 1) = 0$ and $(1, 0) = 0$. Neither is in V , so m-adjacent.
- $(1, 1) \rightarrow (2, 2)$: Shared 4-neighbors are $(1, 2) = 0$ and $(2, 1) = 0$. Neither is in V , so m-adjacent.

Results

The shortest m-path exists and has **length 6**.

Problem 1b) (15 points)

To take a picture for fireworks at night, give at least two operations that should be applied to improve the visualization quality of the image and explain.

Step 1: Understand the Scene Characteristics

Fireworks at night present specific challenges:

- **High dynamic range:** Very bright fireworks against a very dark sky
- **Desired appearance:** Pure black sky with vivid, crisp firework sparks
- **No hidden information:** The dark sky contains nothing meaningful to reveal



Original fireworks image

Step 2: Evaluate Available Operations

We can categorize image processing operations by their effect:

Category A: Operations that brighten dark regions

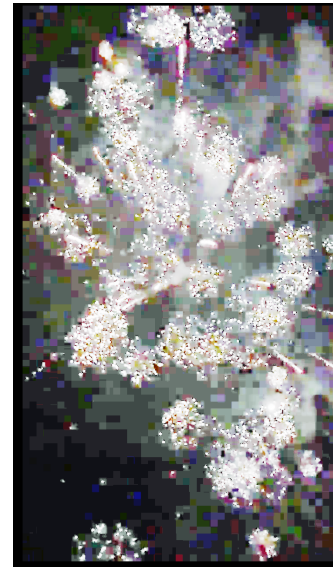
These operations attempt to reveal detail in shadows by lifting dark intensities.



Gamma ($\gamma = 0.5$)



Log Transform

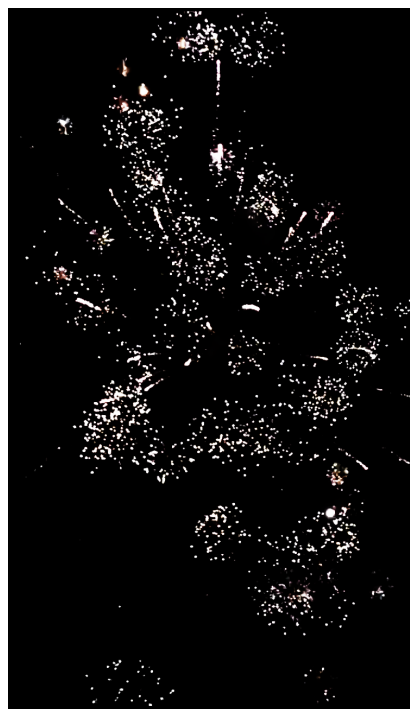


Histogram Equalization

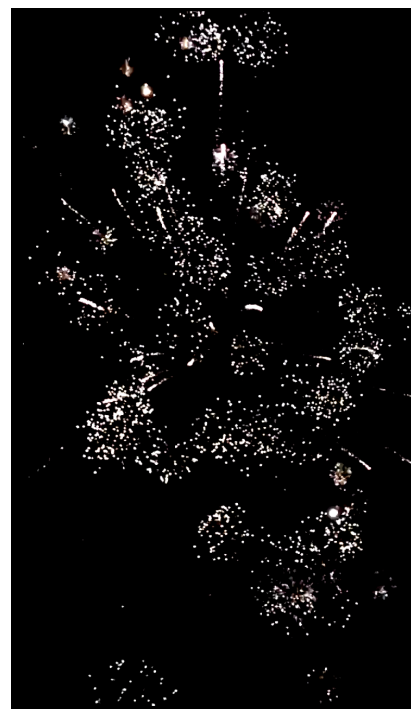
Effect: The black sky becomes gray, revealing noise and sensor artifacts. This is **undesirable** for fireworks because the sky *should* be black—there is no meaningful information to reveal.

Category B: Operations that increase contrast (darks darker, lights lighter)

These operations push dark values toward black and light values toward white.



S-Curve Contrast



Boosted Contrast

Step 3: Consider the Optimization Goal

The “correct” operations depend entirely on what you are trying to achieve:

Goal	Recommended Operations	Avoid
Aesthetic (fireworks)	S-curve, sharpening, saturation	Gamma, Hist EQ
Surveillance (reveal shadows)	Gamma, log transform, Hist EQ	S-curve
Medical imaging	CLAHE, careful contrast	Extreme transforms
Object detection	Thresholding, edge detection	Aesthetic enhancements

The same operation can be *beneficial* or *harmful* depending on the goal. Histogram equalization is excellent for revealing hidden details in surveillance footage, but terrible for fireworks photography where the dark background should remain dark.

Step 4: Consider the Display Medium

The target display device also affects which operations are most beneficial:

Display	Prioritize	Less Important
Mobile phone (small screen)	Color, contrast, brightness	Fine sharpening
Desktop monitor (large screen)	Sharpness, fine detail	Can be subtle
Print	Resolution, color accuracy	Different gamma curve

On a mobile phone, viewers cannot resolve fine detail, so aggressive sharpening provides little benefit—but punchy colors and contrast help images “pop” on a small screen viewed in variable lighting. On a desktop monitor, fine details are visible, making sharpening more valuable.

Results

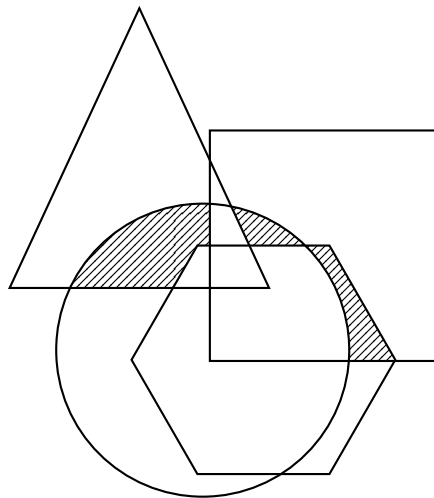
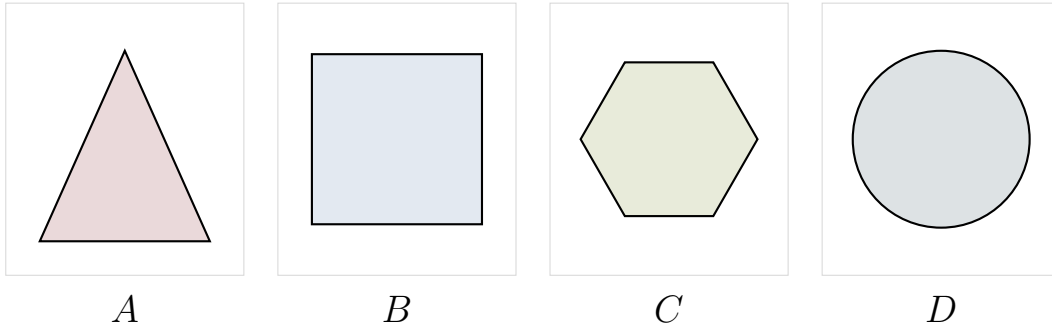
Recommended operations for fireworks photography:

- a) **S-curve contrast enhancement:** Apply a sigmoid transformation $s = \frac{1}{1+e^{-k(r-0.5)}}$ to push dark pixels darker and bright pixels brighter. This ensures a pure black sky while maintaining bright firework sparks.
- b) **Saturation boost** (especially for mobile display): Increase color saturation to make the firework colors more vivid and visually striking.
- c) **Sharpening** (especially for desktop/print): Apply unsharp masking or high-boost filtering to enhance the crispness of firework spark edges.

Operations to avoid: Gamma correction ($\gamma < 1$), log transforms, and histogram equalization—these lift the dark sky and reveal noise, degrading the aesthetic quality.

Problem 2) (20 points)

A, B, C and D are four sets. Give expressions for the set shown shaded in Figure 2 in terms of A, B, C, and D. Note that all of the shaded areas constitute of one set



Problem 3) (25 points)

Given a 3x3 spatial filter $\frac{1}{9} \begin{bmatrix} -1 & -1 & -1 \\ -1 & -17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$ answer the following questions:

- a) (10 points) What type of filter is it? When applying this filter on an image, what kind of effect you will expect in the output image? Why? (Hint: is this filter an image smoothing filter, an image sharpening filter or an edge detector?)

- b) (15 points) Perform the image convolution using this filter on a 3x3 image $\begin{bmatrix} 3 & 5 & 12 \\ 7 & 18 & 11 \\ 5 & 9 & 15 \end{bmatrix}$.

Give the CROPPED filtering result for the output image. (Hint: you need to assume an appropriate zero mapping)

Problem 4) (25 points)

An image with intensities in the range $r \in [0, L - 1]$ has the pdf

$$p_r(r) = \begin{cases} \frac{4r}{3(L-1)^2} & 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

where $L - 1$ is the maximum intensity value. It is desired to transform the intensity levels of this image so that they will have the specified pdf

$$p_z(z) = \begin{cases} \frac{2z}{(L-1)^2} & 0 \leq z \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

Assume that the intensity value is continuous and find the transformation function $z = T(r)$ in terms of r and z that will accomplish this.

(Hints: The transformation function of histogram equalization is given as

$$s = T(r) = (L-1) \int_0^r p_r(w) dw$$

)

Bonus Question (20 points)

Perform the Fourier transform to a function as shown in Figure 3.

(Hint: the definition of the Fourier transform of a function is $F(u) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi ut} dt$)

